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```
clc
clear
close all
```

Problem 1

```
P=[0, 1, 0, 0; 0, 0, 1, 0; 0, 0, 0, 1; 1./2, 1./2, 0, 0] %All states
Aperiodic
n=9;

P_n=P^n
P_10=P^10
P_11=P^11


P=[0, 1, 0, 0; 0, 0, 1, 0; 0, 0, 0, 1; 1./2, 0, 1./2, 0] %All states
Aperiodic

m=3:12;
M=zeros(length(m),1);
power=zeros(length(m),1);
lambda_2=zeros(length(m),1);
lambda_2_mag=zeros(length(m),1);
for k=1:length(m)
power(k)=(m(k)-1).^2+1
P=zeros(m(k),m(k));
P(m(k),1)=1./2;
P(m(k),2)=1./2;
for i=1:(m(k)-1)
     P(i,i+1)=1;
end
lambda=eig(P)
lambda_2(k)=lambda(2);
lambda_2_mag(k)=norm(lambda_2(k));
j=1;
check=0;
while check==0
    if all(P>0,'all');
        P=P;
        check=1;
    else
        P=P*P;
```

```
        j=j+1;
        check=0;
    end
    M(k)=j;
    if j>100
        break
    end

end
end
M
power
lambda_2

T=table(M,power,lambda_2,lambda_2_mag)
writetable(T, 'M_power_lambda_comparision.csv')
```

```
P =

         0    1.0000         0         0
         0         0    1.0000         0
         0         0         0    1.0000
    0.5000    0.5000         0         0


P_n =

         0    0.2500    0.5000    0.2500
    0.1250    0.1250    0.2500    0.5000
    0.2500    0.3750    0.1250    0.2500
    0.1250    0.3750    0.3750    0.1250


P_10 =

    0.1250    0.1250    0.2500    0.5000
    0.2500    0.3750    0.1250    0.2500
    0.1250    0.3750    0.3750    0.1250
    0.0625    0.1875    0.3750    0.3750


P_11 =

    0.2500    0.3750    0.1250    0.2500
    0.1250    0.3750    0.3750    0.1250
    0.0625    0.1875    0.3750    0.3750
    0.1875    0.2500    0.1875    0.3750


P =

         0    1.0000         0         0
         0         0    1.0000         0
```

```
         0         0         0    1.0000
    0.5000         0    0.5000         0


power =

     5
     0
     0
     0
     0
     0
     0
     0
     0
     0


lambda =

  -0.5000 + 0.5000i
  -0.5000 - 0.5000i
   1.0000 + 0.0000i


power =

     5
    10
     0
     0
     0
     0
     0
     0
     0
     0


lambda =

   1.0000 + 0.0000i
  -0.1761 + 0.8607i
  -0.1761 - 0.8607i
  -0.6478 + 0.0000i


power =

     5
    10
    17
     0
     0
```

```
     0
     0
     0
     0
     0


lambda =

   1.0000 + 0.0000i
   0.1688 + 0.9279i
   0.1688 - 0.9279i
  -0.6688 + 0.3389i
  -0.6688 - 0.3389i


power =

     5
    10
    17
    26
     0
     0
     0
     0
     0
     0


lambda =

   1.0000 + 0.0000i
   0.4042 + 0.8809i
   0.4042 - 0.8809i
  -0.5440 + 0.6655i
  -0.5440 - 0.6655i
  -0.7204 + 0.0000i


power =

     5
    10
    17
    26
    37
     0
     0
     0
     0
     0
```

```
lambda =

   1.0000 + 0.0000i
   0.5582 + 0.8073i
   0.5582 - 0.8073i
  -0.3174 + 0.8614i
  -0.3174 - 0.8614i
  -0.7407 + 0.2592i
  -0.7407 - 0.2592i


power =

     5
    10
    17
    26
    37
    50
     0
     0
     0
     0


lambda =

   1.0000 + 0.0000i
   0.6614 + 0.7340i
   0.6614 - 0.7340i
  -0.0954 + 0.9434i
  -0.0954 - 0.9434i
  -0.6834 + 0.5268i
  -0.6834 - 0.5268i
  -0.7651 + 0.0000i


power =

     5
    10
    17
    26
    37
    50
    65
     0
     0
     0


lambda =

   1.0000 + 0.0000i
```

```
   0.7331 + 0.6680i
   0.7331 - 0.6680i
   0.0909 + 0.9609i
   0.0909 - 0.9609i
  -0.5411 + 0.7329i
  -0.5411 - 0.7329i
  -0.7828 + 0.2111i
  -0.7828 - 0.2111i


power =

     5
    10
    17
    26
    37
    50
    65
    82
     0
     0


lambda =

   1.0000 + 0.0000i
   0.7846 + 0.6105i
   0.7846 - 0.6105i
   0.2404 + 0.9453i
   0.2404 - 0.9453i
  -0.3729 + 0.8618i
  -0.3729 - 0.8618i
  -0.7542 + 0.4345i
  -0.7542 - 0.4345i
  -0.7959 + 0.0000i


power =

     5
    10
    17
    26
    37
    50
    65
    82
   101
     0


lambda =
```

```
   1.0000 + 0.0000i
   0.8227 + 0.5608i
   0.8227 - 0.5608i
   0.3593 + 0.9139i
   0.3593 - 0.9139i
  -0.2103 + 0.9328i
  -0.2103 - 0.9328i
  -0.6605 + 0.6267i
  -0.6605 - 0.6267i
  -0.8112 + 0.1786i
  -0.8112 - 0.1786i


power =

     5
    10
    17
    26
    37
    50
    65
    82
   101
   122


lambda =

   1.0000 + 0.0000i
   0.8516 + 0.5178i
   0.8516 - 0.5178i
   0.4541 + 0.8757i
   0.4541 - 0.8757i
  -0.0645 + 0.9653i
  -0.0645 - 0.9653i
  -0.5351 + 0.7672i
  -0.5351 - 0.7672i
  -0.7967 + 0.3698i
  -0.7967 - 0.3698i
  -0.8187 + 0.0000i


M =

     4
     5
     6
     6
     7
     7
     8
     8
     8
```

```
     8


power =

     5
    10
    17
    26
    37
    50
    65
    82
   101
   122


lambda_2 =

  -0.5000 - 0.5000i
  -0.1761 + 0.8607i
   0.1688 + 0.9279i
   0.4042 + 0.8809i
   0.5582 + 0.8073i
   0.6614 + 0.7340i
   0.7331 + 0.6680i
   0.7846 + 0.6105i
   0.8227 + 0.5608i
   0.8516 + 0.5178i


T =

  10×4 table

    M    power         lambda_2          lambda_2_mag
    _    _____    ________________    ____________

    4        5        -0.5-0.5i          0.70711
    5       10      -0.1761+0.86072i     0.87855
    6       17      0.16881+0.92789i     0.94312
    6       26      0.40422+0.88087i     0.96919
    7       37      0.55816+0.80734i      0.9815
    7       50      0.66138+0.73401i     0.98803
    8       65      0.73308+0.66804i     0.99181
    8       82      0.78459+0.61054i     0.99416
    8      101      0.82271+0.56084i     0.99568
    8      122      0.85164+0.51785i     0.99672
```

Problem 2

Read File

```
T=readtable("big12.csv");
season=T(:,1);
week=T(:,2);
date=T(:,3);
home=T(:,4);
home_pts=T(:,5);
away=T(:,6);
away_pts=T(:,7);
neutral=T(:,8);
comleted=T(:,9);

T_2025=T(T.season==2025,:);
T_2026=T(T.season==2026,:);

teams = {'Oklahoma State','Texas Tech','Kansas State','Houston', ...
         'Arizona','Utah','Kansas','Baylor','Arizona State', ...
         'Iowa State','Colorado','BYU','Cincinnati','UCF', ...
         'TCU','West Virginia'};
%{
for i=1:length(teams)
     T_temp=T_2025(T_2025.home==teams(i) || T_2025.away==teams(i),:);
end
%}

P=[0,1,0,0;0,0,1,0;0,0,0,1;1/2,0,1/2,0];
e=eig(P)
P=[0,1,0,0,0,0;0,0,1,0,0,0;0,0,0,1,0,0;0,0,0,0,1,0;0,0,0,0,0,1;1/2,0,1/2,0,0,
0];
e=eig(P)


e =

  -1.0000 + 0.0000i
   1.0000 + 0.0000i
  -0.0000 + 0.7071i
  -0.0000 - 0.7071i


e =

  -1.0000 + 0.0000i
   1.0000 + 0.0000i
  -0.3218 + 0.7769i
  -0.3218 - 0.7769i
   0.3218 + 0.7769i
   0.3218 - 0.7769i
```

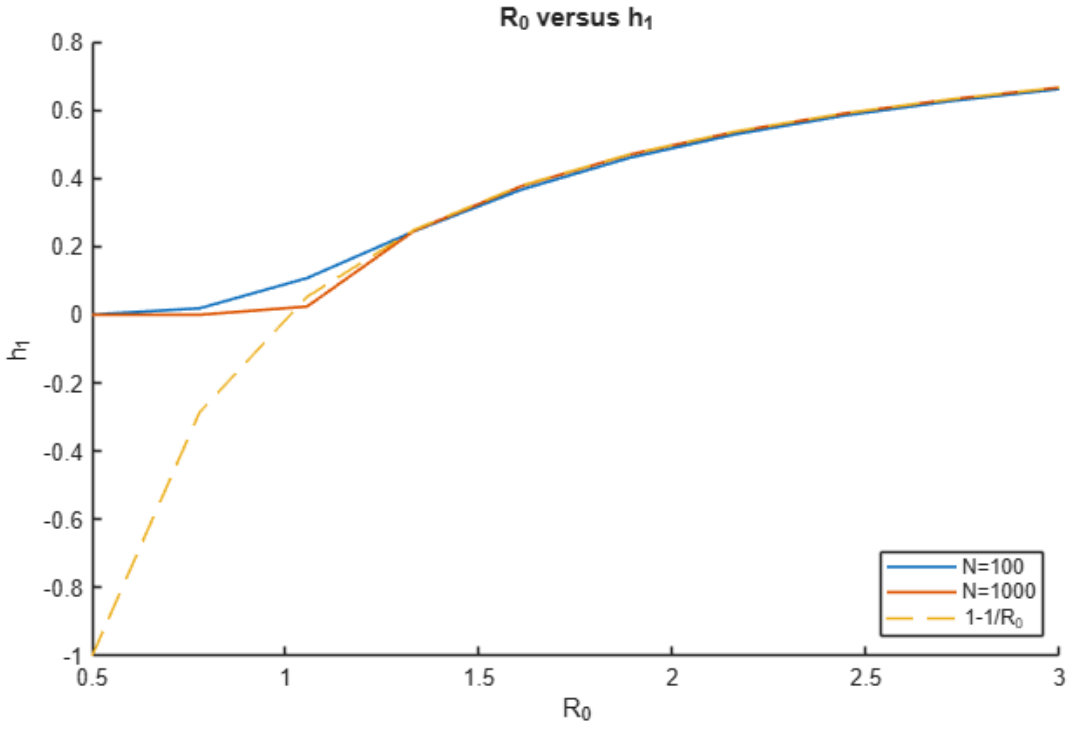
Problem 3

```
%c
N=[100; 1000];
```

```
K_c=N./10;
R_0_c=linspace(.5,3,10);


for k=1:length(R_0_c)
rho(1)=1;
for i=1:(K_c(1)-1)
    rho(i+1)=rho(i).*N(1)./(R_0_c(k).*(N(1)-(i)));
end
rho_100(k)=sum(rho);

rho_1(1)=1;
for i=1:(K_c(2)-1)
    rho_1(i+1)=rho_1(i).*N(2)./(R_0_c(k).*(N(2)-(i)));
end
rho_1000(k)=sum(rho_1);
end
h_1_100=1./rho_100;
h_1_1000=1./rho_1000;
figure
hold on
plot(R_0_c,h_1_100,'LineWidth',1.2)
plot(R_0_c,h_1_1000,'LineWidth',1.2)
plot(R_0_c,1-1./R_0_c,'--')
legend('N=100','N=1000','1-1/R_0','Location','best')
xlabel('R_0')
ylabel('h_1')
title('R_0 versus h_1')
hold off
```



d

```
N=100;
K=10;
gamma=1;
t=.005;
r_0=[.5;.8;1.5;3];
beta=r_0.*gamma./N;

P=zeros(N,N);
P(1,1)=1-beta(1).*(N-1).*t;
P(1,2)=beta(1).*(N-1).*t;
P(N,N)=1-gamma*t;
P(N,N-1)=gamma^t;
e(1)=1+P(1,2);
for i=2:(N-2)
p=beta(1).*i.*(N-i).*t;
q=gamma.*i.*t;

P(i,i+1)=p;
P(i,i-1)=q;
P(i,i)=1-p-q;
e(i)=1-q+p;
end
e(N)=1-P(N,N-1);
E=table(e)
writetable(E, 'Expectation_simulation.csv')
```

```
E =

  100×1 table

       e
    _______

     1.0025
     0.9949
    0.99228
     0.9896
    0.98687
     0.9841
    0.98128
     0.9784
    0.97547
     0.9725
    0.96947
     0.9664
    0.96328
     0.9601
    0.95688
     0.9536
    0.95027
```

```
 0.9469
0.94348
   0.94
0.93648
 0.9329
0.92927
 0.9256
0.92188
 0.9181
0.91427
 0.9104
0.90648
 0.9025
0.89848
 0.8944
0.89027
 0.8861
0.88187
 0.8776
0.87327
 0.8689
0.86447
   0.86
0.85547
 0.8509
0.84627
 0.8416
0.83688
 0.8321
0.82727
 0.8224
0.81747
 0.8125
0.80748
 0.8024
0.79727
 0.7921
0.78687
 0.7816
0.77627
 0.7709
0.76548
   0.76
0.75448
 0.7489
0.74328
 0.7376
0.73188
 0.7261
0.72027
 0.7144
0.70848
 0.7025
0.69648
```

```
 0.6904
0.68427
 0.6781
0.67188
 0.6656
0.65927
 0.6529
0.64648
   0.64
0.63348
 0.6269
0.62027
 0.6136
0.60687
 0.6001
0.59327
 0.5864
0.57947
 0.5725
0.56547
 0.5584
0.55127
 0.5441
0.53687
 0.5296
0.52228
 0.5149
      0
      0
```

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